

Product Line Profitability & Margin Performance Analysis

Nassau Candy Distributor - Internal Analytics Report

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Prepared for Nassau Candy Distributor leadership

SECTION 1 - Executive Summary

- Total revenue is \$141,783.63 with total gross profit of \$93,442.80 and a company-wide weighted margin of 65.91%.
- The top 5 products, representing 33.3% of SKUs, generate at least 80% of total profit.
- Chocolate contributes 95.06% of gross profit, creating division-level concentration risk.
- Wonka Bar - Scrumdiddlyumptious is the highest profit product at \$19,357.50.
- Kazookles are flagged for discontinuation review based on low margin and low profit contribution.

KPI	Value
Total Revenue	\$141,783.63
Total Gross Profit	\$93,442.80
Avg Margin %	65.91%
SKU Count	15

The portfolio is profitable overall, but leadership should not manage it through sales volume alone. Profit concentration in Chocolate masks operational weakness in selected Other-division SKUs, especially products with high cost-to-revenue ratios. The immediate management opportunity is to protect the leading Wonka Bar economics while repricing, renegotiating cost, or retiring structurally weak SKUs.

SECTION 2 - Background & Problem Statement

Sales volume alone is misleading for Nassau Candy because high-revenue products can still dilute earnings when cost structures are unfavorable. Revenue also obscures concentration risk: a product line can appear healthy while a small number of products or one division carries nearly all profit.

- Which products create the largest gross profit pool rather than just the largest sales pool.
- Which products have weak margin economics and should be repriced, renegotiated, or reviewed for discontinuation.
- How profit is concentrated across products, divisions, and factories.
- Whether factory and division performance is efficient enough to support growth plans.

The analysis covers the three active product divisions: Chocolate, Sugar, and Other. It also evaluates the five factories present in the cleaned data: Lot's O' Nuts, Wicked Choccy's, Secret Factory, The Other Factory, and Sugar Shack.

SECTION 3 - Methodology

- Loaded the source transaction file and profiled shape, data types, and null counts.
- Validated Sales, Cost, Units, and Gross Profit; removed invalid non-positive Sales or Units records.
- Imputed numeric missing values with medians and removed rows missing required product or division labels.
- Removed exact duplicates, parsed dates, standardized division labels, and normalized product names.
- Mapped each product and division to its factory and calculated profitability KPI fields.

KPI formulas used: $\text{Gross Margin \%} = \text{Gross Profit} / \text{Sales} \times 100$; $\text{Profit Per Unit} = \text{Gross Profit} / \text{Units}$; $\text{Revenue Contribution \%} = \text{Product Sales} / \text{Total Sales} \times 100$; $\text{Cost-to-Revenue Ratio} = \text{Cost} / \text{Sales}$.

Validation included gross profit reconciliation against Sales minus Cost with a \$0.01 tolerance, post-cleaning null checks, duplicate checks, and factory mapping coverage checks.

SECTION 4 - Product-Level Findings

Top 5 products by gross profit:

Product	Division	Sales	Gross Profit	Margin
Wonka Bar - Scrumdiddlyumptious	Chocolate	\$27,874.80	\$19,357.50	69.44%
Wonka Bar - Triple Dazzle Caramel	Chocolate	\$28,485.00	\$18,610.20	65.33%
Wonka Bar - Milk Chocolate	Chocolate	\$26,867.75	\$17,443.37	64.92%
Wonka Bar - Nutty Crunch Surprise	Chocolate	\$23,574.95	\$16,819.95	71.35%
Wonka Bar - Fudge Mallows	Chocolate	\$24,890.40	\$16,593.60	66.67%

Bottom 5 products by margin percentage:

Product	Division	Sales	Gross Profit	Margin
Kazookles	Other	\$1,205.75	\$92.75	7.69%
Fun Dip	Sugar	\$12.00	\$4.80	40.00%
SweeTARTS	Sugar	\$61.50	\$28.70	46.67%
Nerds	Sugar	\$15.00	\$7.00	46.67%
Lickable Wallpaper	Other	\$7,860.00	\$3,930.00	50.00%

Quadrant	SKU Count
Star	5
Cash Drain	3
Niche	3
Deadweight	4

Products flagged for discontinuation review:

Product	Margin	Profit Share	Cost Ratio
Kazookles	7.69%	0.10%	92.31%

SECTION 5 - Division Performance Analysis

Division	Revenue	Profit	Margin	Profit Share	Flag
Chocolate	\$131,692.90	\$88,824.62	67.45%	95.06%	Efficient
Other	\$9,742.00	\$4,380.70	44.97%	4.69%	Efficient
Sugar	\$348.73	\$237.48	68.10%	0.25%	Efficient

Factory	Profit	Margin	SKU Count
Lot's O' Nuts	\$52,771.05	69.13%	3
Wicked Choccy's	\$36,053.57	65.13%	2
Secret Factory	\$4,344.70	50.59%	3
The Other Factory	\$152.25	11.87%	2
Sugar Shack	\$121.23	54.86%	5

Chocolate: Revenue is \$131,692.90, gross profit is \$88,824.62, and weighted margin is 67.45%. The division contributes 95.06% of profit and is classified as efficient.

Other: Revenue is \$9,742.00, gross profit is \$4,380.70, and weighted margin is 44.97%. The division contributes 4.69% of profit and is classified as efficient.

Sugar: Revenue is \$348.73, gross profit is \$237.48, and weighted margin is 68.10%. The division contributes 0.25% of profit and is classified as efficient.

SECTION 6 - Pareto & Concentration Risk

5 products (33.3% of SKUs) generate 80% of total profit. The first product crossing the 80% threshold is Wonka Bar - Fudge Mallows.

- Single-product dependency risk above 30% of profit: None.
- Division dependency risk above 60% of profit: Chocolate.
- Reduce concentration risk by building attach strategies for Sugar and Other products with strong margins but low volume.
- Protect the leading Chocolate portfolio through price discipline and supply continuity planning.

SECTION 7 - Cost Structure Diagnostics

Cost-heavy products: Kazookles.

Pricing inefficiency flags: Kazookles.

Action	Recommended SKUs
Repricing	Kazookles, Fun Dip, SweeTARTS, Nerds, Lickable Wallpaper
Cost renegotiation	Kazookles, Fun Dip, SweeTARTS, Nerds, Lickable Wallpaper
Discontinuation	Kazookles

SECTION 8 - Recommendations

Immediate actions - this quarter:

1. Open a discontinuation and pricing review for Kazookles; expected impact: remove or repair structurally weak margin drag.
2. Renegotiate cost inputs for Kazookles, Fun Dip, SweetARTS; expected impact: lower cost-to-revenue exposure in the weakest SKUs.
3. Protect Chocolate pricing and inventory availability; expected impact: preserve the division currently funding most company profit.

Medium-term initiatives - 6 months:

1. Bundle high-margin, low-volume Sugar and Other SKUs with leading Chocolate products; expected impact: diversify profit without sacrificing margin quality.
2. Build monthly product profitability monitoring with alerts for margin below 20% or cost ratio above 75%; expected impact: faster intervention.
3. Review factory-level production economics for The Other Factory; expected impact: identify supplier or production changes for low-margin output.

Strategic review items - annual:

1. Set target profit contribution by division; expected impact: reduce over-dependency on Chocolate.
2. Conduct SKU rationalization using quadrant status and Pareto contribution; expected impact: focus management attention on profit-relevant lines.
3. Evaluate whether low-volume niche SKUs support strategic assortment goals; expected impact: clarify which products deserve investment despite small revenue.

SECTION 9 - Conclusion

Nassau Candy Distributor has a profitable portfolio, but the economics are uneven. Chocolate products carry most of the profit base, while several smaller SKUs contribute little to enterprise earnings. The company should therefore manage profitability at the product and factory level, not only at the division revenue level.

The analysis shows that a small number of products drive the majority of profit. This is operationally efficient in the short run, but it creates concentration exposure if demand, supplier cost, or pricing power changes in the leading Chocolate line.

Leadership should prioritize margin repair on weak SKUs, protect high-profit Chocolate products, and develop a more balanced growth path through selected Sugar and Other products. The recommended next step is to convert this report into a recurring monthly profitability review with action owners for pricing, sourcing, and product portfolio decisions.

Closing statement: Nassau Candy can improve resilience and profitability by shifting from sales-led reporting to margin-led product management.